The Bottom of the Shub–Smale Tau Conjecture: an Exact Census of Integer Roots for Constant-Free Straight-Line Programs of Length at Most Seven

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Abstract

For a univariate integer polynomial f, let $\tau(f)$ be the minimum number of $+, -, \times$ gates needed to compute f from x and the constants $-1, 0, 1$, and let $z(f)$ be its number of distinct integer roots. The Shub–Smale tau conjecture (Smale's fourth problem) asserts $z(f) \leq (1 + \tau(f))^\kappa$ for a universal κ; it implies $\mathrm{P}_\mathbb{C} \neq \mathrm{NP}_\mathbb{C}$ in the Blum–Shub–Smale model and $\mathrm{VP}^0 \neq \mathrm{VNP}^0$, and it is open even for $\kappa = 1$. We report the first exact census of the conjecture's growth function at the bottom of its ladder: writing $z_{\max}(\tau)$ for the maximum of $z(f)$ over all nonzero f with $\tau(f) \leq \tau$, we prove by exhaustive, exactly verified computation that

$$z_{\max}(1), \ldots, z_{\max}(7) \;=\; 1,\ 2,\ 3,\ 3,\ 4,\ 5,\ 5.$$

In particular the minimum cost of 4 distinct integer roots is 5 gates, of 5 roots is 6 gates, and of 6 roots is 8 or 9 gates; the growth function has plateaus at $\tau = 4$ and $\tau = 7$, so an extra gate does not always buy an extra root. The censuses are decision-complete: depth 6 required the exact construction of all 25,844,905 reachable computation states, and depth 7 a complete scan of 2,013,706 new polynomials, using a "last-gate" lemma that decides one depth beyond any exhausted frontier without storing the next frontier. The enumerator is anchored on Markström's published exhaustive census of integer targets (all fourteen anchor values reproduced exactly) and cross-checked against an independent computer-algebra root counter on 284 polynomials. We further prove two elementary stall theorems explaining the observed record mechanisms: for any monic $h \in \mathbb{Z}[x]$ of degree at least 2, towers built by iterating h alone have depth-independent integer root counts (for $h = x^2 - 2$, the iterates $h^{\circ k}(x) - x$ keep exactly 2 integer roots against 2^k real roots), and across the quadratic family $h_c = x^2 - c$ with $c \leq 200$ the maximum tower yield is 5, attained only at $c = 2$; a second yield series at $c = m^2 + m + 1$, produced by genuine integer 2-cycles, is identified and closed by the classical fact that integer polynomial cycles have length at most 2. All computations are exact integer arithmetic, reproducible from committed code with hypotheses declared before each run; two of our own pre-registered predictions were refuted by the machine and are reported as such. This is an experimental-mathematics record in the tradition of Markström's census: it decides nothing about the conjecture asymptotically, and says so.

1 Introduction

1.1 The conjecture

A *constant-free straight-line program* (SLP) over $\mathbb{Z}[x]$ is a sequence $(1, x, f_2, \ldots, f_N)$ in which every f_i ($i \geq 2$) is a sum, difference, or product of two earlier entries; following [1, 8] we write $\tau(f)$ for the least $N-1$ over all such programs with $f_N = f$. Equivalently [6], $\tau(f)$ is the minimal number of operation gates of an arithmetic circuit over the inputs x and the free constants $-1, 0, 1$. Let $z(f)$ denote the number of distinct roots of f in $\mathbb{Z}$.

Conjecture 1.1 (Shub–Smale [1]; Smale's fourth problem [2]). *There is a universal constant $\kappa \geq 1$ such that every nonzero $f \in \mathbb{Z}[x]$ satisfies $z(f) \leq (1+\tau(f))^\kappa$.*

The conjecture fails for $\kappa < 1$: the geometric-progression products $(x-2)(x-4)\cdots(x-2^{2^j})$ acquire one root per $O(1)$ gates [8]. It is open even for $\kappa = 1$. Its significance is structural: Shub and Smale proved that it implies $\mathrm{P}_\mathbb{C} \neq \mathrm{NP}_\mathbb{C}$ in the Blum–Shub–Smale model [1, 3], the proof reducing to the hardness of computing all sequences $(m_n \cdot n!)$; Bürgisser [5], building on Koiran [4], proved that it implies $\mathrm{VP}^0 \neq \mathrm{VNP}^0$, and that the single sequence $(n!)$ being hard to compute already yields that separation. Easy factorials would give nonuniform polynomial-time integer factoring [3, 9]; with integer division allowed, $n!$ *is* easy [11], so the restriction to $+, -, \times$ carries the entire content. The real-zeros analogue of the conjecture is false: iterating the logistic map produces 2^j real roots at cost $O(j)$ [8], which motivated Koiran's real tau conjecture for structured expressions [13, 14, 15, 16, 17]. On the lower-bound side, no nontrivial lower bound on $\tau(n!)$ is known [4]; on the reduction side, Rojas proved that a p-adic digit version of the conjecture, for any fixed prime, already implies the full statement, via a p-independent subquadratic bound on the valuation spectrum of roots [8]. Recent work applies the conjecture to explicit exponential lower bounds [18] and tightens the bridge from BSS separations to constant-free Valiant separations [19]; the current survey reference is [6, §4.6].

1.2 What this paper does

The literature contains, to our knowledge, exactly one exhaustive computation in this model: Markström's census of *integer targets* (the straight-line complexity of small factorials and primorials), complete to program length 11 on 2013-era hardware [7]. No published computation addresses the *polynomial* side, i.e. the actual growth function of the conjecture. We define

$$z_{\max}(\tau) \;=\; \max\{\, z(f) \;:\; f \in \mathbb{Z}[x] \setminus \{0\},\; \tau(f) \leq \tau \,\}$$

and determine it exactly for $\tau \leq 7$, together with the full record galleries, explicit optimal witnesses, and the exact cost thresholds for $3, 4, 5$ and (within one gate) 6 distinct integer roots. We also prove two elementary theorems that explain the observed record mechanisms and close entire construction classes (Section 6), and we measure the quadratic tower family exhaustively (Section 7).

Our contributions:

1. **The census** (Section 4): $z_{\max}(1..7) = 1, 2, 3, 3, 4, 5, 5$, decision-complete, with per-depth state and polynomial counts, record galleries and replayed witness programs. Plateaus occur at $\tau = 4$ and $\tau = 7$.

2. **Exact thresholds**: the minimum τ admitting 3 distinct integer roots is 3 ($x^3 - x$); for 4 roots it is 5; for 5 roots it is 6; for 6 roots it lies in $[8, 9]$ (the census excludes 7; a 9-gate witness exists).

3. **A method lemma** (Lemma 2.5): every polynomial of cost $d + 1$ is one gate over a normalized depth-d state, so $z_{\max}(d+1)$ is computable from an exhausted depth-d frontier without storing the depth-$(d + 1)$ frontier. This bought depth 7 (a $\sim 10^9$-state frontier avoided).

4. **Stall theorems** (Section 6): iterating any fixed monic map of degree ≥ 2 yields depth-independent integer root counts; for the Chebyshev-conjugate map x^2-2, whose difference-of-squares splittings generate all census records at depths 5–7, the exact stable core is $\{0, \pm 1, \pm 2\}$.

5. **The family measurement** (Section 7): across $h_c = x^2 - c$, $c \leq 200$, and all tower shapes to four iterations, the maximum integer-root yield is 5, attained only at $c = 2$; two yield-4 series exist, $c = m(m + 1)$ (fixed and anti-fixed points) and $c = m^2 + m + 1$ (genuine 2-cycles), and the classical bound on integer polynomial cycle lengths shows this inventory is complete.

6. **Anchoring and audit** (Section 3): the enumerator reproduces all fourteen of Markström's published anchor values; an independent computer-algebra cross-check agrees on 284 polynomials; every experiment's hypothesis, budget and kill criterion were committed before its run, and two of our pre-registered predictions were refuted by the machine (both disclosed below).

Every claim in this paper is labeled: **[MV]** machine-verified in exact arithmetic from committed code, **[D]** proved here (elementary, self-contained), or **[C]** conjectural/observational. Nothing here decides the conjecture in either direction, and no asymptotic consequence is claimed: the census maps where the conjecture lives at the bottom of its ladder, in the same genre as [7].

2 Model, normalization, and the census method

Definition 2.1. Programs operate on values in $\mathbb{Z}[x]$. The inputs are -1, 1 and x; each gate applies $+$, $-$ or $\times$ to two (not necessarily distinct) earlier values; the length of a program is its number of gates; $\tau(f)$ is the minimal length of a program computing f at some gate.

Lemma 2.2 (free 0 is redundant)**.** *Adding* 0 *as a fourth free input does not change* τ*.* [D]

Proof. In any program using the input 0: $a + 0$ and $a - 0$ duplicate a; $a \times 0$ computes 0; $0 - a$ equals $(-1) \times a$, one gate either way. Replacing each use and deleting duplicated gates never lengthens the program. □

Lemma 2.3 (normalization)**.** *Every f has an optimal program in which no gate recomputes an input or an earlier value, and no gate computes* 0*.* [D]

Proof. A gate duplicating an available value can be deleted, rewiring its uses to the original; a computed 0 is usable only in the redundant patterns of Lemma 2.2. Each deletion shortens the program, so an optimal one contains neither. □

Lemma 2.4 (reached-set sufficiency)**.** *The extensions available to a program depend only on the* set *of values it has computed. Consequently a breadth-first search over reached sets ("states"), with set-level deduplication, visits every normalized reachable state of each depth exactly once.* [D]

This is the polynomial analogue of the range-isomorphism reduction of [7]. The census algorithm is exactly this BFS, in exact integer arithmetic on dense coefficient tuples, with each new polynomial's first appearance depth recorded (that depth is $\tau(f)$, by Lemmas 2.3 and 2.4). Integer roots are counted exactly: strip the power of x, then test the divisors of the trailing coefficient (both signs).

Lemma 2.5 (last gate)**.** *Let $\mathcal{F}_d$ be the set of normalized reached states of depth d. Every f with $\tau(f) = d+1$ is $a \circ b$ for a single gate $\circ$ and operands a, b available in some state of $\mathcal{F}_d$. Hence $z_{\max}(d+1) = \max\big(z_{\max}(d),\ \max z(a \circ b)\big)$ over one op over every state of $\mathcal{F}_d$, and this maximum is computable without constructing $\mathcal{F}_{d+1}$.* [D]

Proof. Take an optimal program of length $d+1$ for f and normalize it (Lemma 2.3; deletions would shorten it, contradicting optimality, so it is already normalized). Its first d gates form a normalized reached state of depth d, and its last gate is one op over that state's available values. □

3 Anchoring and audit discipline

Markström anchor [MV]. Restricted to the integer model (start value 1, positive normalized values), our enumerator reproduces Figure 1 of [7] exactly at every compared depth: reached-set sizes $2, 4, 9, 26, 102, 562, 4363$ and complete initial intervals $2, 4, 6, 12, 40, 112, 310$ for lengths 1 through 7: fourteen of fourteen anchor values.

Independent cross-check [MV]. An independent root counter (SymPy's exact factorization) agrees with ours on all 284 polynomials checked: every polynomial of $\tau \leq 3$ and every record polynomial of the censuses.

Engine re-derivation [MV]. The depth-7 computation uses a reimplemented, memoized engine; before its new output was trusted, it was required to re-derive every prior census value (state counts 9/98/1462/29506/778087, new-polynomial counts through depth 6, and the record sets), and did.

Pre-registration. Every computation ran under a hypothesis file committed to the repository *before* the run, with falsifiable predictions, an explicit statement of what a pass and a fail would prove, a compute budget and a kill criterion. Two committed predictions were refuted by the machine: we predicted $z_{\max}(6) = 4$ (the census returned 5; Section 4), and we predicted zero tower yield off the fixed-point series (a second series exists; Section 7). Both refutations are reported alongside what they taught. One tooling incident is likewise on record: divisor-based root counting is infeasible against tower constant terms of size c^{2^k}, and was replaced, for the family measurement only, by direct evaluation over a window proved correct in advance (Section 7), cross-checked against the divisor method where both run.

4 The census

Theorem 4.1 (the census; MV)**.** *In the model of Definition 2.1,*

τ	1	2	3	4	5	6	7
$z_{\max}(\tau)$	1	2	3	3	4	5	5

and the censuses behind each value are decision-complete.

The supporting exhaustive counts are:

τ	1	2	3	4	5	6	7
reached states	9	98	1,462	29,506	778,087	25,844,905	(not stored)
new polynomials	9	34	177	1,249	11,377	134,494	2,013,706
$z_{\max}$	1	2	3	3	4	5	5

Depth 7 was decided by Lemma 2.5 over the exhausted depth-6 frontier; among its 2,013,706 new polynomials the distinct-integer-root histogram is 742,635 rootless, 904,421 with one, 299,488 with two, 63,903 with three, 3,196 with four, 63 with five, and none with six or more.

Theorem 4.2 (cost thresholds; MV)**.** *The minimum τ admitting k distinct integer roots is: $k = 1\colon 1$; $k = 2\colon 2$; $k = 3\colon 3$; $k = 4\colon 5$; $k = 5\colon 6$; and for $k = 6$ it lies in $\{8, 9\}$.*

The $k = 6$ upper bound is the 9-gate witness $q(q-2)(q-6)$ with $q = x^2 - x$ (root set $\{-2, -1, 0, 1, 2, 3\}$); the lower bound is Theorem 4.1. Representative optimal witnesses, replayed gate by gate in exact arithmetic from the committed artifacts:

τ	record	witness program
3	$x^3 - x$, roots $\{0, \pm 1\}$	$x{\cdot}x$; $x^2{-}1$; $x{\cdot}(x^2{-}1)$
4	$-x(x{+}1)(x{+}2)$, roots $\{0, -1, -2\}$	$-1{-}x$; $1{-}(-1{-}x)$; $x(-1{-}x)$; $(x^2{+}x)\ldots$
5	$x^2 - (x^2{-}2)^2 = -(x^2{-}1)(x^2{-}4)$	-2; x^2; $x^2{-}2$; $(x^2{-}2)^2$; $x^2 - (x^2{-}2)^2$
6	$\mp x(x^2{-}1)(x^2{-}4)$, roots $\{0, \pm 1, \pm 2\}$	the $\tau{=}5$ record, then $\times\, x$

Plateaus. $z_{\max}$ pauses at $\tau = 4$ and $\tau = 7$. The record galleries explain the asymmetry: the step $5 \to 6$ succeeds because multiplying by the input x costs one gate and adjoins the root 0 to a record avoiding it; the steps $3 \to 4$ and $6 \to 7$ fail because every candidate next root beyond the "free" set $\{0, \pm 1, \pm 2\}$ requires a built constant, and constants cost gates. The $\tau = 5$ records (ten polynomials) and $\tau = 6$ records (four) are all difference-of-squares splittings over the map $x^2 - 2$ (Section 6); at $\tau = 7$ the sixty-three five-rooters appear, but no six-rooter.

5 Record mechanisms

The census records decompose into a small move inventory [MV for the instances; observational as a classification]: linear factors $x - a$ for available constants a; block translation at one gate per unit shift; multiplication by the input x (one gate, adjoins 0); and difference-of-squares splittings $B^2 - A^2 = (B - A)(B + A)$ at two gates given A and B. The depth-5 record $x^2 - (x^2 - 2)^2 = -(x{-}1)(x{+}1)(x{-}2)(x{+}2)$ is the fundamental instance: its inner map $C(x) = x^2 - 2$ is the Chebyshev-conjugate doubling map ($h(z) = z + 1/z$ satisfies $h(z)^2 - 2 = h(z^2)$), i.e. the record is the integer shadow of exactly the iteration that makes the *real* analogue of the conjecture false. Over $\mathbb{Z}$, that factory is sterile beyond one splitting, which the next section proves.

6 Stall theorems

Lemma 6.1 (escape)**.** *Let $h = x^d + \sum_{i<d} a_i x^i \in \mathbb{Z}[x]$, $d \geq 2$, and $R = 1 + \sum_{i<d} |a_i|$. If $|x| \geq R+1$ then $|h(x)| \geq |x|+1$.* [D]

Proof. $\sum_{i<d} |a_i|\,|x|^i \leq (R-1)|x|^{d-1}$ for $|x| \geq 1$, so $|h(x)| \geq |x|^{d-1}(|x| - (R-1)) \geq 2|x|^{d-1} \geq 2|x| \geq |x|+1$ for $|x| \geq R+1$. □

Theorem 6.2 (monic stall; D)**.** *Fix a monic $h \in \mathbb{Z}[x]$ of degree ≥ 2. There is a constant $Z(h)$ such that for every $k \geq 1$, both $h^{\circ k}(x) - x$ and the difference-of-squares towers $G_k = h^{\circ(k-1)}(x)^2 - h^{\circ k}(x)^2$ have at most $Z(h)$ distinct integer roots. One may take $Z(h) = \#([-M, M] \cap \mathbb{Z})$ where $M = \max(R+1, \max |a|)$ over the integer solutions a of $h(y) = \pm y$.*

Proof. Integer roots of $h^{\circ k}(x) - x$ are periodic points of h; by Lemma 6.1 all integer periodic points lie in $[-R-1, R+1]$. For G_k, factor $G_k = (h^{\circ(k-1)} - h^{\circ k})(h^{\circ(k-1)} + h^{\circ k})$: an integer root x has $y = h^{\circ(k-1)}(x)$ solving $h(y) = \pm y$, so x lies in the increasing union $\bigcup_j (h^{\circ j})^{-1}(A) \cap \mathbb{Z}$ for the finite set A of such y. If $|x| > M$ then $|h^{\circ j}(x)| \geq |x| + j > \max_{a \in A} |a|$ for all j by Lemma 6.1 and induction, so no element of the union exceeds M in absolute value; the union is contained in a fixed finite set and stabilizes. □

Proposition 6.3 (the Chebyshev core; D, MV spot-checks)**.** *For $C(x) = x^2 - 2$: the integer periodic points are exactly $\{2, -1\}$; for every $k \geq 1$ the polynomial $C^{\circ k}(x) - x$ has exactly 2 distinct integer roots while having 2^k distinct real roots and $\tau \leq 2k+2$; and the towers G_k have root set $\{\pm 1, \pm 2\}$ for $k = 1$ and exactly $\{0, \pm 1, \pm 2\}$ for every $k \geq 2$.*

Proof. Orbits: $0 \mapsto -2 \mapsto 2 \mapsto 2$ and $1 \mapsto -1 \mapsto -1$; by Lemma 6.1 (with $R = 3$; direct check for $|x| \geq 3$) all other integer orbits escape, so the only cycles are the fixed points 2 and -1, giving the first two claims (the real-root count is the classical Chebyshev conjugacy). For G_k: $h(y) = y$ has integer solutions $\{2, -1\}$ and $h(y) = -y$ has $\{-2, 1\}$; integer preimages are $C^{-1}(2) = \{\pm 2\}$, $C^{-1}(-1) = \{\pm 1\}$, $C^{-1}(-2) = \{0\}$, $C^{-1}(1) = C^{-1}(0) = \emptyset$, so the preimage closure of $\{\pm 1, \pm 2\}$ is $\{0, \pm 1, \pm 2\}$ and is stable. □

Theorem 6.2 closes a construction class: no family built by iterating one fixed monic map can witness superpolynomial z versus τ; a refutation of the conjecture, if one exists, must build fresh constants or mix maps, paying gates for them. That is precisely the regime the census measures.

7 The quadratic family

We measured all tower shapes ($h^{\circ k}(x) - x$ and $h^{\circ i}(x)^2 - h^{\circ j}(x)^2$, $0 \leq i < j \leq 4$) across $h_c = x^2 - c$ for $1 \leq c \leq 200$, in exact arithmetic. Root finding here cannot use trailing-coefficient divisors (those constants reach $c^{2^k} \sim 10^{18}$); instead, by Lemma 6.1, every integer root of such a shape lies in $[-(c+1), c+1]$, and direct evaluation over that window is exact. The two finders agree on every shape where both are feasible ($c \leq 10$, $j \leq 2$) [MV].

Theorem 7.1 (family census; MV)**.** *For $1 \leq c \leq 200$ and the shapes above: the maximum number of distinct integer roots is 5, attained only at $c = 2$ (the set $\{0, \pm 1, \pm 2\}$ of Proposition 6.3). Nonzero yields occur exactly on two series: $c = m(m+1)$, yield 4 for $m \geq 2$ with root set $\{\pm m, \pm(m+1)\}$ (from the fixed points $\{m+1, -m\}$ and anti-fixed points*

$\{m, -m-1\}$), and $c = m^2 + m + 1$, yield 4 with the same root sets, produced by the genuine 2-cycles $m \mapsto -m-1 \mapsto m$ (for $c = 1$: yield 3, root set $\{0, \pm 1\}$, from the cycle $0 \leftrightarrow -1$).

The second series refuted our committed prediction that only the fixed-point series yields roots. It is, however, the *last* such surprise available to this construction class:

Proposition 7.2 (cycle ceiling; D, classical)**.** *For any $f \in \mathbb{Z}[x]$, every cycle of f in $\mathbb{Z}$ has length at most 2. Consequently fixed points, anti-fixed points and 2-cycles are the complete periodic inventory harvestable by tower shapes over any integer polynomial map.*

Proof. For integers $a \neq b$, $(a - b) \mid (f(a) - f(b))$. Around a cycle $a_0 \mapsto a_1 \mapsto \cdots \mapsto a_r = a_0$ of distinct points, the differences $d_i = a_{i+1} - a_i$ satisfy $d_i \mid d_{i+1}$ cyclically, so all $|d_i|$ equal one value $d > 0$, and the cycle is a closed walk on the arithmetic line $a_0 + d\,\mathbb{Z}$ with steps $\pm d$. If two consecutive steps have opposite signs then $a_{i+2} = a_i$, which forces $r = 2$; if all steps have the same sign the walk never closes. Hence $r \leq 2$. The fact is classical in the polynomial-cycles literature (see Narkiewicz [21]); the proof is included to keep the paper self-contained. □

Yield per gate across the family is maximized at $c = 2$ and decreases in c (the yield is bounded by 5 while building c costs $\tau(c) \sim \log c$ gates, measured exactly against our anchored integer census): within this family, the "parameterized loophole" left open by Theorem 6.2 is empty [MV for $c \leq 200$; C beyond].

8 Limits, and what would count as progress

Nothing above bears on the conjecture asymptotically: censuses to depth 7 cannot distinguish polynomial from superpolynomial growth, and our theorems close construction classes rather than open the bounding problem. The honest state of the measured landscape is: every mechanism observed or excluded so far pays $\Theta(1)$ gates per root, the growth function's first structure (plateaus at 4 and 7) is the visible cost of constant-building, and the geometric-progression family remains the best known factory, linear-rate, exactly as when Rojas noted that the conjecture fails for $\kappa < 1$ [8].

Concrete open items this record makes precise: (i) close the $\{8, 9\}$ window of Theorem 4.2: the 8-gate decision is finite and SAT-shaped (in the style of exact SLP synthesis over GF(2) [20]); (ii) extend the integer frontier past Markström's length 11 [7], including his monotonicity question for $\tau(n!)$; (iii) measure the growth of the valuation-spectrum bound $s \leq N_p(s) \leq s(s+1)/2$ of [8], where the true rate is stated as open; (iv) the census's 2-adic instrumentation records that all depth-5–7 record polynomials concentrate their roots on the valuations $\{0, 1\}$ [MV], consistent with the pressure sitting on the digit-conjecture side of Rojas' reduction [C].

Data and code availability

All code, hypotheses, run artifacts, verdicts and the wiki are committed at `github.com/fsantibanezleal/CAOS_R` under `problems/computation-complexity/tau-conjecture/` (experiments `EXP-001`–`EXP-005`; package `tclib` with its test suite). All computations are exact integer arithmetic in Python 3.13 standard library; total compute for every result in this paper is under two hours on one desktop CPU (24 cores available, single-threaded runs; peak memory ∼6 GB).

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